Airbag-Based Protective System for UAV Medical Payloads

Design and Simulation in LS-DYNA

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by

Annamaria Iannuzzo

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Preface

A preface...

Annamaria Iannuzzo
Delft, October 2026

Summary

A summary...

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Nomenclature

If a nomenclature is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Abbreviations

Abbreviation	Definition
ISA	International Standard Atmosphere
...	

Symbols

Symbol	Definition	Unit
V	Velocity	[m/s]
...		
ρ	Density	[kg/m^3]
...		

1

Introduction

1.1. Motivation

The present MSc Thesis in Aerospace Structures and Materials falls under the category of *Engineering Design*, in alternative to the *Scientific Research*-type projects. In *design* projects, it is required to attempt the process of innovation not through the discovery and exploration of a research niche, but rather by applying pre-existing scientific knowledge to enable the concrete realisation of a technical design. In the case of the present thesis, the overall objective consists in proposing a *deisign workflow* and validating it through testing and simulation. If adopted in the preliminary design phase of a drone-based delivery mission, this workflow yields an optimised airbag-type device tailored to a specific failure-and-drop application, in line with the requirements set by the project commissioners. Therefore, the successful completion of this thesis is achieved by:

1. Proposing an application-specific design workflow that fits within the broader context of the ongoing research effort;
2. Testing the workflow, both experimentally and through simulation, to realistically inform the device's design.

The design workflow (informally called *inner (design) loop* throughout this thesis) is inteded as a first attempt to bridge the gap between the long-term, broadly-defined objective of constructing a large-scale drone network (also called *outer (design) loop*) over the region of Piedmont, Italy, and the previously completed research work, consisting in a series of logistic, legislative, and generally high-level evaluations of possible **drone-based missions** with the objective to **deliver medical supplies in emergency scenarios**. Such work was conducted *Politecnico di Torino*, in collaboration with the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex and automotive consulting company *FEV Italia S.r.l.*, three of the main industrial and research stakeholders working towards helping hospitals in the transition from road to air transport to enhance the logistics of the *Sistema Sanitario Nazionale* (*SSN*, the Italian National Healthcare System) both at an intra- and inter-hospital level.

This MSc Thesis Project is, therefore, a multidisciplinary project where the knowledge and techniques of Structural Engineering are put to the service of the medical field. The airbag-like device resulting from the design process should be able to protect the free-falling medical payload in the event of a drone failure and its subsequent detachment from the latter. The device aims to bring the impact accelerations within the allowable limits of the payload container and of the medical payload being transported, in compliance with the safety regulations regarding dispersion of chemical substances into the environment. However, contrary to most aeronautical projects which do not distinguish between the integrity of the aerial vehicle and the safety of its passengers, this project adopts a "payload-first" approach where the protection of the highly valuable medical cargo is prioritised over the integrity of the drone carrying it. Distinguishing between the safety aspects of the unmanned aerial vehicle (UAV) and its payload implies a certain facilitation in the certification process of the UAV, as well as for the modular implementation of the safety device on a variety of UAVs serving in different, specific delivery

missions.

1.2. Project framing

1.2.1. Care centralisation and territorial inequity in the Italian healthcare system

The idea of transforming healthcare delivery networks from road- to air-transport-based systems stems from the consensus within the Italian medical field, consolidated during the public health emergency of the COVID-19 pandemic (2020–2022), that the current ambulance-based architecture of the *SSN*'s logistic infrastructure (described in detail[1] in [1]) cannot accommodate the healthcare system's needs and transitions. The two main - and competing - factors are:

1. **Care centralisation**: [2] describes care centralisation as the ongoing process of concentrating medical expertise and services in key geographical areas, e.g. provincial and/or regional capitals, to meet the demands of high-volume care. The downside of care centralisation is that at a regional level, it induces "distance-related inequity, particularly for rural and peripheral populations", especially heightened in emergency scenarios.
2. **Unhomogeneous distribution of emergency services** : [3] affirms that, in contrast with the centralisation manouver described above, emergency resources are still unhomogeneously distributed over the regional territories, with services overloaded in densely populated areas and underused in low-density ones, resulting in inefficient overall utilisation.

In conclusion, [4] builds on these two points by affirming the importance of establishing a generally more efficient strategy for healthcare management at all levels, and incentivises the development of technologies suitable for upgrading, among the several aspects of the *SSN*, the logistics of the Italian healthcare supply chain.

1.2.2. Prior research and positioning of the present thesis

In response to the issues highlighted in Section 1.2.1, recent years have seen an increasing number of technology development initiatives where industry players in the transport sector and research institutions work together to push for the modernisation of transportation means within the Italian *SSN*. Several of these industry-research partnerships have ventured into the globally booming market of unmanned aerial vehicles, where remotely-piloted or fully autonomous drones are being used ever more often to carry medical payloads in a plethora of scenarios: intra- vs. inter-hospital delivery [5], ordinary logistics vs. emergency missions and humanitarian applications[6], experimental/direct flight between strategic poles [7] vs. a fully-operational, inter-hospital, hub-based UAV network covering national territory (the *Zipline* case study[8]).

Following the global trend to answer a local need, in the specific case of the Piedmont region in north-west Italy, where at the time of writing there exists no city- or region-wide UAV network deployed for the delivery of medical goods, many pilot programmes (further analysed in Section 2.1) are emerging, among which a partnership between the Pharmacy department of the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex (head-quartered at Molinette Hospital in Turin), a research group within the Department of Mechanical and Aerospace Engineering at *Politecnico di Torino*, and the Numerical Anaysis Team of *FEV Italia S.r.l.*, the Italian branch of the German-founded multinational company *FEV*, a global leader in research and development for the automotive industry.

The six-year-long partnership between these three institutions is the backbone of the present MSc thesis project, the most recent of a chain of research projects aiming to pursue the long-term vision of flanking, if not completely replacing, the current road-based transportation means with aerial alternatives within hospitals in Piedmont and, specifically, in the city of Turin (the region's capital), in favour of more rapid, more efficient, and safer transportation of medical supplies. Specifically, this research effort has broadly tackled the key aspects required to establish a complete and regulation-compliant UAV medical delivery ecosystem in the Piedmontese territory, resulting in five MSc theses and one conference paper presenting the following results:

[1] *"The type of vehicles used varies from region to region. In Piedmont it mainly consists of ambulances used only for transport plus more advanced vehicles equipped for resuscitation, replaced in some sites by mSa on a car with a doctor, nurse and rescuer driver, organized on the basis of specific needs of the territory, different in the four macro-areas of the emergency network (rEmE) in which the region was divided."*[1]

- **Cavallo (2020)** [9]: Preliminary assessment of UAV platforms suitable under Italian and European regulations. Development of risk maps and definition of optimal trajectories between selected hospitals.
- **Bertelli (2021)** [10]: Detailed selection of UAV based on mission specifications derived from the medical goods to be transported and ENAC regulatory constraints. Application of the SORA (Specific Operations Risk Assessment) methodology to three feasible mission routes, yielding the safety requirements that operators must satisfy.
- **Sivieri (2021)** [11]: Identification and characterisation of thermal and vibrational loads encountered during typical UAV medical missions. Experimental testing on milk-derived sera samples to determine their resistance to UAV-induced vibrational stresses. First hypothesis of medical payload container geometry and materials, implemented in the experimental campaign.
- **Fiorentino (2021)** [12]: Development of a path-planning strategy for life-saving drug distribution in populated environments. Generation of risk maps and implementation of an RRT-based algorithm to minimise global risk and flight time.
- **Angelini (2023)** [13] and **Angelini (2024)** [14]: Comparative analysis of the safety and economic efficiency of emergency medical deliveries performed by car, drone, and VTOL aircraft. Identification of optimal pharmaceutical delivery hub locations for each transport mode to maximise logistical performance.

The present thesis was carried out in collaboration with the three institutions mentioned above to advance the research effort, bridging the gap between their long-term objective and the results achieved so far. This required preliminary work aimed at formalising the project requirements at both a broad, high level and at a specific, first-step level. Having taken a legislative, logistic, market- and mission-evaluation approach until now, it was possible to delineate a scenario in which the approval of *ENAC* (the Italian Civil Aviation Authority - *Ente Nazionale per l'Aviazione Civile*) could potentially be obtained to enable this type of operation. However, as certifications are unavoidably specific to mission type, trajectory and vehicle, it appeared necessary to implement a more concrete approach rooted in *design*, whilst also adopting a "payload-first" approach that diverted the project towards a device for the *safety of the medical payload* rather, than that of the aerial vehicle. The deliberate prioritisation of this aspect is justified in Sections 2.1 and 2.2, where it is illustrated on the basis of a regional market analysis and the exploration of an ongoing paradigm shift in the design process of biomedical devices.

1.2.3. Stakeholders

Overall, the stakeholders participating in the present project had the following roles:

- **TU Delft** (Prof. Bianca Giovanardi): Primary academic supervisor overseeing the scientific quality and methodological development of the research.
- **Politecnico di Torino** (Prof. Enrico Cestino, Davide Angelini): Institutional leads and co-supervisors for the broader UAV medical-delivery research project, providing access to the experimental platforms used.
- **A.O.U. Città della Salute e della Scienza di Torino / Molinette Hospital**: Provider of medical and operational data related to pharmaceutical transport requirements and mission scenarios.
- **FEV Italia S.r.l.**: Industrial partner offering technical support for advanced modelling and simulation activities, including SolidWorks-based design and LS-DYNA impact analyses.

The interaction between stakeholders, as well as the management and communication aspect of the project, is analysed in Chapter 6 - "Project Management".

1.3. Document Outline

DISCORSIVO. Esempio:

The thesis is structured as follows, first, chapter 2 presents a literature review to outline the several possibilities to predict the buckling load of cylindrical shells and which techniques are available to reduce the sensitivity to imperfections. The chapter concludes by identifying the research gaps related to the buckling of cylindrical shells. From this conclusion, the research objective and scope are defined

in chapter 3. The goal is to include imperfections within the displacement-based Koiter methodology and the necessary steps to achieve this are explained in chapter 4 to 7, as well as how effective this novel modelling technique is. Finally, the concluding remarks can be found in chapter 8 and future research is presented in chapter 9.

2

Background and Research-Development Gap

In technology development, it is important to gain a clear idea of two aspects:

1. **Research aspect**: if the technology that is being developed requires the exploration of an uncharted research niche, as opposed to utilising pre-existing and consolidated knowledge for the creation of a new product;
2. **Development aspect**: whether the proposed technology targets a well-defined market with unmet customer demand and no established supply.

This chapter explores both these aspects in Sections 2.2-2.3 and 2.1, respectively. The aim is to gather sufficient background information, from market allocation of the proposed technology to concrete understanding of airbag design criteria, to depict the choices proposed in Chapter 3 regarding operative *methodology* as an answer to the Research Questions proposed at the end of this chapter.

2.1. Regional Market Analysis: medical UAV logistics in Piedmont

The Italian National Healthcare System (*SSN*) is structured hierarchically: the guidelines and regulations issued at a *national* level must be implemented locally, at a *regional* level, by each region's *Sistema Sanitario Regionale* (*SSR*, the Regional Healthcare Service). Vice versa, aviation rules follow European-level enforcement mechanisms: Following the 2019 harmonisation of European drone operations across the "open" and "specific" categories [15], operators seeking authorisation for high-risk flights are required to conduct a thorough risk analysis using the *Specific Operations Risk Assessment (SORA)* methodology developed by JARUS[1], making a detailed evaluation of potential failure scenarios while proving compliance with the demanding safety standards imposed by aviation authorities.

The discrepancy between modi operandi of the aviation and healthcare authorities has shaped the way drone-network operations have mostly been developing in the last decades in Italy: several emerging, albeit scattered, research groups, either cooperating or competing with industrial counterparts, have achieved geographically circumscribed trial deployments while being unable to scale the operation beyond testing.

A complete summary of such operations, limited to the means of investigation available for the present thesis project, can be found in Table A.2 in Appendix A.2. The table shows a clear trend: although potentially booming as several healthcare facilities are exploring the possibility of adopting drone transport and operators are proposing new drone models for them, most activities stop after the initial demonstrations, with the only exception being the region of Abruzzo where the operational phase of the drone drug-delivery service between a strategic hospital and village pharmacies was announced in July 2025 [16], [17]. In general, a pattern of fragmentation, short-lived pilots, and few transitions

[1] *Joint Authorities for Rulemaking on Unmanned Systems*

to a potentially scalable service is observed in the construction of the overall *value chain*, with many independent institutions covering equivalent roles, as shown in Table A.1 in Appendix A.1.

Table 2.1: Actors involved in drone delivery of medical goods in Piedmont, grouped by project.

Project	Role	Institution	Scope
INDOOR (2021-2024)	Promoter	Fondazione DOT	Project promoter [18]
	Funder	Centro Nazionale Trapianti	Research grant at PIC4SeR [19]
	Funder	City of Turin	Project partner [18]
	Research	PIC4SeR, Politecnico di Torino	Autonomous flight technologies for transporting biological material [20]
	Research	Università di Torino	Project partner [18]
	Technology partner	Mavtech	Technical partner; aerial surveillance and operational-support products [18]
	Operator, payload maker	ABzero (Pisa)	Supplied the drone and the Smart Capsule (sensorised container for blood, organs and drugs); handled the ENAC authorisation [19], [21], [22], [23]
	End user	AOU Città della Salute e della Scienza	Test flight between hospital sites [22]
	End user	Regional Transplant Centre Piemonte–Valle d'Aosta	Organ and biological transport [18]
	Regulator	ENAC	National civil aviation authority; project partner and flight authorisation [18]
DIMEAS feasibility studies (2020-2024)	Research	DIMEAS, Politecnico di Torino (Prof. E. Cestino)	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost–safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Technology partner	FEV Italia	Industrial partner; numerical impact analysis
	End user	AOU Città della Salute e della Scienza	Pharmacy department partner
Hydrogen logistics plan	End user	AOU Città della Salute e della Scienza	Proposed hydrogen drone logistics plan [24]
	Infrastructure	Torino Caselle and Cuneo Levaldigi airports	Proposed hub and support base [24]
LastMed	Funder	City of Turin	Funded the LastMed call [25]
	Alternative (ground)	Tecnocad / Tecnomill / Bylogix	Autonomous ground rover for last-mile drug delivery [26]
Independent	Drone manufacturer	SkyEU	Autonomous high-speed eVTOL drones for first aid and critical logistics [27]
	Regulator	Regione Piemonte	Regional health authority; drone trials subject to benefit–risk analysis [28]

Adopting a region-based, rather than a nation-based, approach for market analysis is, therefore, necessary. Table 2.1 highlights stakeholders involved in Piedmont-based operations and their roles in the value chain, illustrating a fragmented scenario: Project INDOOR, launched by Fondazione DOT in 2021, culminated in a 500 m-long test flight by an autonomous UAV between two Città della Salute hospital sites, carrying medical goods in a dedicated capsule, on 1st October 2024. In this project, several other stakeholders were involved (in fact, most of the ones present in Table 2.1), among which ABzero (a spin-off from *Scuola Normale Superiore di Pisa* responsible for capsule design and handling the ENAC authorisation process for the test flight), parallel research group PIC4SeR from the Department of Electronics and Telecommunications at *Politecnico di Torino*, and *MAVTech S.r.l.* (UAV systems designer company based in Turin). Independently of the INDOOR project of 2021-2024, the startup SkyEU was founded in 2025 promising to develop eVTOL drone systems for critical logistics within, but not limited to, the healthcare sector.

In parallel, and independently of INDOOR, the research group led by Prof. Enrico Cestino at the Department of Mechanical and Aerospace Engineering (DIMEAS) of *Politecnico di Torino*, together with *A.O.U. Città della Salute e della Scienza* and *FEV Italia*, has pursued since 2020 a series of feasibility studies on the Molinette-San Giovanni Bosco route (Section 1.2.2). These covered regulatory and mission evaluation, risk mapping and path planning, payload environmental loads, and cost-safety comparisons between road, drone and eVTOL transport. No flight campaign has been carried out so far. This makes the group the only Piedmontese actor whose work so far has been entirely at the planning and assessment level.

On the other hand, the regional administration's general stance with respect to this type of operation highlights the fundamental requirement of thorough benefit-risk analyses to guarantee both the safety of the Piedmontese citizens and the preservation of the medical cargo, possibly needing to limit transportation to exceptional emergencies in mountain areas [28]. From an engineering perspective, these criticalities can be translated in the following barriers for the development of a full-scale, UAV-based delivery network:

1. **Authorisation of flights**: Operations beyond the open category are approved by compiling the SORA. Among the different classes of mitigations, *ground risk* mitigations (one of the first steps required in the SORA) reduce the consequences of the UAV's impact on the ground after a loss of control, for example by introducing a parachute. It is key for technical stakeholders to prove the reliability of their implemented ground risk mitigations to protect civilians in case of drone failure.
2. **Payload integrity and containment**: In accordance with the regional administration's concerns, current certification standard UN3373 for ground handling of Biological Substances, Category B, establishes that packages must pass a drop test at 1.2 m with no leakage from the primary container CITE STANDARD HERE. However, in the event of a free-fall from ordinary mission altitudes, this standard loses its significance and the potential for biological material dispersion into the environment becomes highly relevant.
3. **Unsustainability of the current business models** While costs of road and helicopter transport of biomedical goods such as organs, drugs and other critical supplies are expected to fall significantly with drones, most trials are funded by EU projects and one-time grants, with the consequence of making the project unsustainable and non-scalable.

On the basis of this analysis, several research and development gaps can be identified: SkyEU is endeavouring to enact an economically sustainable business model for large-scale UAV networks, but has a small public record; ABzero has patented capsules equipped with electronics that reliably monitor in-flight vibrations and guarantee thermal control of the medical payload, but these are only certified to the UN3373 standard, whereas in case of in-flight failure a much more conservative approach is needed; regional authorities need reassurance on the safety aspect for civilians, as a 500 m test flight on a controlled route shows that the obstacle is not drone-making, but guaranteeing safety in everyday performance rather than in single monitored missions.

In this scenario, the institutions (*Politecnico di Torino (DIMEAS), FEV Italia, AOU Città della Salute*) that commissioned the present thesis project aim to tackle, among the three barriers, barrier 2, concerning payload integrity and containment. The proposed solution is an airbag-type device that protects the payload and prevents its dispersion in the event of an impact, especially in view of transporting extremely valuable medicinals and/or biological materials, e.g. human organs for urgent transplant operations. This device is conceived as an add-on to existing capsules such as the one by ABzero, mounted on drones such as SkyEU's, and is intended to help close the certification gap highlighted above, while acknowledging that it does not address the issues linked to SORA ground risk or the sustainability of the business model.

2.2. Mission-first vs. Safety-first: a design-philosophy gap

This section builds on the research-development gap presented earlier by attempting to connect its root causes to a discrepancy between design philosophies applied in the aeronautical and the biomedical sectors. The proposed design solution (the airbag-type device) is, in fact, a suggestion to solve an "aeronautical problem" implementing a "medical approach".

The issue of proving compliance to match demanding safety standards imposed by the certification authorities does not appear to be a Piedmont-only (or even an Italy-only) problem, but rather a widespread trend that affects all non-standard aeronautical projects across the field. As highlighted, for example, in [29], most UAV producers focus on optimising their aerial vehicle's design, minimising mission-related vibrational loads, thermal loads, overall mass, but also proposing ever more efficient cargo-drop mechanisms and ways to overcome geographical obstacles, all partaking in a feedback loop against aerodynamic and flight-mechanics performance metrics. This machine-centric approach appears to be the default, especially considering that UAV-based missions are, naturally, radically different from standard airline and/or general aviation operations, where passenger safety and vehicle safety are inherently aligned and where, therefore, stringent certification requirements strongly limit design freedom CITE ARTICLE HERE. A common consequence of focusing exclusively on technological potential when designing UAV-based operations is, therefore, neglecting the safety assurance aspect of the project, which is usually delegated to including a parachute-type device in the final design phases. As pointed out by [30] in their design feasibility evaluation study, the introduction of a parachute usually increases the risk of falling into a mass-increase feedback loop: the parachute increases the airframe's overall mass, which then requires a larger, thus heavier, parachute for efficient deceleration during free-fall, which again results in an excessive mass increase, and so on. This negative feedback loop results either in never reaching design convergence, or otherwise in having to accept a smaller parachute with reduced deceleration capacity, still implying a relatively high ground-impact velocity and almost certainly destructive damage to the airframe, its cargo, and the impacted areas on the ground.

The general impression given by the brief market analysis conducted earlier and the observations gathered in the present paragraph is that, perhaps, the "mission-first" approach adopted in the majority of nascent UAV-based operations is not sustainable for the projects' longevity beyond pilot flights, and that it might be necessary to embrace a paradigm shift in design approaches. Specifically, there might be potential in drawing inspiration from the biomedical field, where medical technology designers are encouraged to shift from a techno-centric to a human-centred approach to avoid the *translation gap* or *implementation gap*, i.e. the potential to introduce or enhance health inequity when trying to implement extremely advanced technology that does not take into account all necessities and limitations of human patients [31].

Just like success is measured on the patient and not on the device in the biomedical field, in the case of UAV-based medical delivery operations success should be measured on payload integrity and protection of civilians, not (only) on the capabilities of the aerial vehicle. A drone that survives an in-flight failure while sacrificing its payload and threatening irreversible human damage has undoubtedly failed its mission, as for a biomedical device that functions correctly but does not serve the patient. This reasoning leads to the first of the contributions proposed in the present thesis, which is the *"safety-first"* design principle presented in Section 3.2.1. This thesis aims to serve as a test-bed for this proposed approach, where exploring the airbag design process helps bridge the gap between current development efforts and certification.

2.3. Airbag impact attenuation: background knowledge

Airbag-type devices have a well-established, enduring research-development record with contributions found across multiple industries. The information presented in Sections 2.3.1 and 2.3.2 aims to represent the State of the Art practices adopted in airbag design. The scope of this investigation is developing a solid, although perhaps not comprehensive, understanding of design recommendations on which to base the proposal for the design workflow of later Section 3.2.2.

2.3.1. Established applications and common working principle

Airbag technology is well established across two primary sectors: *automotive* and *aerospace*.

In the automotive industry, these systems are implemented during vehicle crashes, distinguishing between general automobile crashes and motorcycle impacts. In the space industry, ever since the early space missions of the 1960s, airbags have been deployed during the final landing phases of space capsules. Within the aeronautical industry, specifically in the "specific" operations category, other niche applications have emerged, such as the experimental study of adaptive airbag systems for the protection of UAVs and general aviation vehicles [32] (following a long track record of exploration in the field of

adaptive technologies for impact absorption [33], [34]).

In either case, the primary objective of the airbag-type device remains the same: to be stored compactly in ordinary mission conditions, inflate within the allowable time before impact (whether during an emergency impact or landing), and decelerate the target body (whether a crash test dummy or a spacecraft) far more gently than in case of direct impact. When analysing these systems, purely analytical, hard-codable solutions are typically limited to single-degree-of-freedom (1-DoF) formulations, immediately resolving to the use of far more computationally-heavy finite element analyses to capture the complex two- and three-dimensional effects of the impact. The only exception to this trend is the technical report NATICK/TR-87/042 published by T. F. Patterson in 1985[2], which presents a 6-DoF analytical simulation built on closed-form component models, but remains inaccessible at the time of writing of the present literature search.

Despite the diversity of fields, all applications share identical fundamental working principles. An example of an analytical study of airbag-based impact events, and one of the most significant papers in shaping this thesis' work, is Zhou et al. (2019) [35], in turn heavily rooted in the SAND93-2133 technical report [36] on a code for predicting the performance of the Pathfinder lander's airbag impact attenuation system, which describes the core working principle of airbag physics in a one-degree-of-freedom impact event.

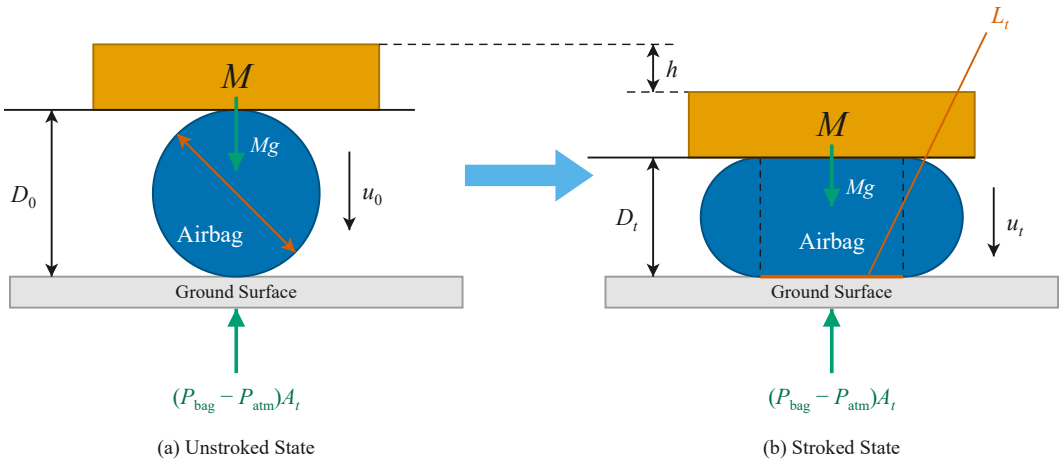


Figure 2.1: One-degree-of-freedom airbag impact model in the unstroked (a) and stroked (b) states, adapted from Zhou et al. [35].

The quantities under study for this type of phenomenon are shown in Fig.2.1, where a point load (yet, with a finite extension, as to avoid penetration in the airbag and justify the crude, though analytically necessary, *uniform pressure/control volume* assumption, where pressure is uniformly distributed within the airbag volume) is protected by a single inflated airbag in incipient contact with the ground surface after a free fall. The quantities are:

- M, the mass of the payload, which remains constant throughout the analysis and results in the vertically applied gravitational force Mg.
- u_0, the initial impact velocity at time t_0, which evolves to u_t after each subsequent time increment. It is updated at every time increment through kinematic analysis.
- D_0, the initial diameter of the airbag, which evolves to D_t following the same loop update concept. Broadly speaking, its meaning can be extended to the characteristic dimension of the airbag, according to the airbag shape selected for the test case among the available options (vertically- or horizontally-oriented cylinder, sphere, toroid, etc.).
- h, the *stroke*, i.e. the (vertical) distance covered by the payload when compressing the airbag during the impact event.

[2] T. F. Patterson, *Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers*, NATICK/TR-87/042, U.S. Army Natick Research, Development and Engineering Center, Aug. 1985.

- $(P_{bag} - P_{atm}) \cdot A_t$, the normal reaction force from the ground, based on the difference between the internal pressure of the airbag P_{bag} and atmospheric pressure P_{atm}, multiplied by the footprint area A_t with the ground, calculated according to different geometric assumptions correlating $A_t = f(h, t)$.

The reference system is downward-positive, in accordance with the direction of gravity and the initial velocity, making the reaction force negative.

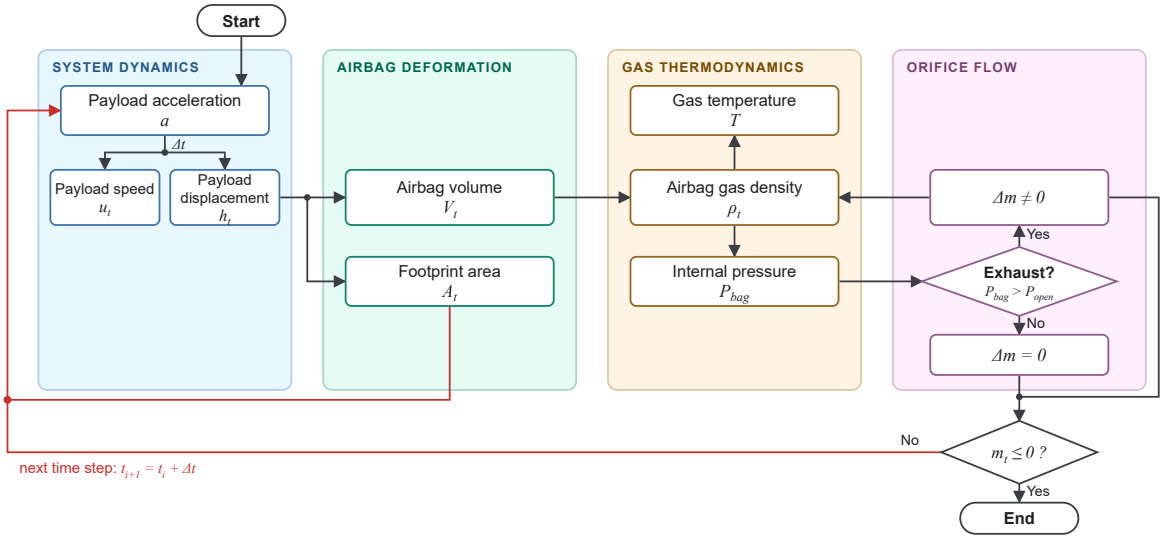


Figure 2.2: Calculation flowchart of the one-degree-of-freedom airbag impact model, adapted from Zhou et al. [35].

Except for the initial conditions and the payload's mass, these quantities are updated iteratively at each timestep to produce characteristic outputs for airbag-governed impact events such as *airbag volume*, *internal pressure*, *internal gas mass*, *payload acceleration*, *payload speed*, and *payload displacement*. The logic of this calculation loop is schematized in Fig.2.2, which is an adaptation of the original diagram presented in the paper. The calculation loop is divided into four blocks, run sequentially at each timestep:

1. **System dynamics step**: The resultant acceleration a acting on the payload is calculated based on the reaction force from the ground calculated in the previous time step, and then integrated using the *central difference scheme* to obtain payload velocity u_t and displacement (or stroke) h.
2. **Airbag deformation step**: The airbag volume V_t and footprint area A_t on the ground surface are updated according to the kinematic information from the previous timestep. In this step, the hypothesis linking the stroke to the footprint area is enforced based on geometric assumptions, as mentioned earlier.
3. **Gas thermodynamics step**: The state variables of the gas are updated assuming adiabatic compression, as it is safe to assume that little to no heat is exchanged during the rapid compression process.
4. **Orifice flow step**: This critical step accounts for energy dissipation during the impact event. By modelling the outward airflow as flow through a one-way valve, treated as an orifice embedded in an impermeable fabric, typical nozzle fluid dynamics equations are applied to calculate the Δm vented at each timestep. In the case of this specific paper, mass is vented only if the internal pressure P_{bag} exceeds threshold pressure P_{open}, greater than atmospheric pressure P_{amb}. Within a single timestep, venting takes place as long as the exhaust condition withstands, updating gas density ρ_t and recalculating the gas' temperature T and pressure in an internal feedback loop.

In [35]'s framing, the code stops either when the manually-set duration of the phenomenon is reached, or when all mass is vented off. Testing of the code indicates that the no-rebound condition shown in the paper is numerically enforced by introducing an if-check managing the application of the reaction force.

The core work of the present thesis resides in turning the working principle of airbag impact events

as presented in [35] into an informed, multi-variable problem that requires numerical optimisation. The code presented in later Section 4.3 is an expanded version of this implementation, incorporating additional criteria and alternative termination mechanisms.

2.3.2. Design guidelines, requirements, and variables

Framing

In essence, an airbag can be considered as an energy absorber and dissipator that must bring the payload from its impact velocity v to rest within an available stroke S, without exceeding an allowable deceleration a. This relation is expressed in its most basic form by the kinematic relation [37], [38]

$$S = \frac{v^2}{2a} \tag{2.1}$$

which shows the central trade-off in airbag design: the lower the allowable deceleration, the longer the stroke needed to decelerate the payload and, therefore, the larger the airbag. During compression, the airbag behaves as a non-linear spring whose stiffness depends on its geometry and venting characteristics [35], [36], [39]. Venting is what dissipates the absorbed energy [39]; whatever energy remains stored in the gas at the end of the stroke is returned to the payload as rebound [40]. Increasing the volume of the airbag and inflation pressure raises its energy dissipation capacity, but at the cost of higher membrane loads on the fabric [37], [39].

Configuration concepts

Airbags are generally classified into three types depending on whether they contemplate the presence of venting in their design: *non-vented*, *vented*, and *combined* [35]. A combined airbag usually consists of an outer, main-venting airbag and an inner, non-venting, anti-bottoming airbag. This "bag within a bag" concept prevents the payload from bottoming out once the outer bag has vented [39], [41], [42], and commonly relies on complex sequential inflation of the outer and inner bags [42]. An alternative to this design is using multiple airbags side by side rather than one within the other, following a "fail-safe" design philosophy by sharing impact loads and providing redundancy in case of airbag failure [37]. Rather than completely enveloping the payload, the airbag can act as a buffer cushion beneath it [42], or inflate between plates attached to the sides of the cargo [43]. Moreover, venting can be passive, activated pneumatically, or active, activated electronically [34], [40]. Active venting is the basis of adaptive impact absorption, in which the airbag responds to the actual, case-specific impact load [32]: for example, in multi-chamber designs, controllable valves allow gas to flow between adjacent chambers and to the environment, redistributing a non-uniform load across the structure [33], [34].

Design requirements

Notable airbag design requirements and recommendations identified in literature are summarised in Table 2.2 and relate to impact conditions, such as the vertical and horizontal velocity, impact attitude, and landing surface [37], [39], [44]; to the allowable deceleration of the payload; and to general constraints on mass, packaging volume and deployment time [32], [42], [45].

Table 2.2: Design requirements for airbag landing and impact-attenuation systems identified in the literature.

Category	Requirement	Sources
Dynamics	Limit the peak acceleration transmitted to the payload	[32], [39], [42], [46], [47]
	Reduce the touchdown velocity to a target value	[32]
	Cope with horizontal velocity (non-nominal impact attitudes coupling)	[37], [39], [44]
Landing conditions	Cope with different landing surfaces	[41], [44], [48]
	Avoid fabric rupture and tearing	[39], [45]
	Prevent bottoming-out of the payload	[35], [39], [41], [42]
	Limit rebound and prevent rollover	[40], [42]
Inflation pre-landing	Synchronise venting and payload contact	[49]

continued on next page

Category	Requirement	Sources
	Deploy fully within the available time	[32], [42]
	Retain pressure for the required duration	[42], [45]
Packaging	Survive long-term stowage in the packed state	[45]
	Integrate with the host vehicle without interference	[32]
	Withstand the environmental temperature range	[44], [45]
General	Minimise system mass	[32], [42], [45]
	Allow reuse	[42], [44]
	Provide redundancy against single-bag failure	[37]

Design variables

All airbag design variables found in the literature analysed for the present review can be found in Table 2.3. The variables in the table are divided into two umbrella categories: *Energy absorption at impact* and *Deployment and packaging*. These categories can be mapped almost directly onto two distinct impact phases (at-impact and pre-impact behaviour, respectively) and, consequently, into two separate, although not entirely uncoupled, design stages: *phase 1*, focusing on the inflated configuration during impact to *limit peak acceleration* below the allowable limit set by the payload's integrity requirement, and *Phase 2*, focusing on the pre-impact inflation process by mostly *optimising out-of-stowage deployment time* and leading to prototyping.

Table 2.3: Airbag design variables identified in the literature, grouped by design focus and subcategory, with the associated trade-offs.

Group	Variable	Trade-off	Sources
Energy absorption at impact			
Geometry	Shape	Pressure-vessel shapes carry membrane loads well; faceted shapes are easier to build	[35], [37], [39], [41], [48], [50]
	Characteristic size	More volume absorbs more energy but raises membrane loads, chance of rebound, and overall structural weight	[37], [39], [40], [51]
	Payload contact area	Larger contact spreads the load; wrapping the entire payload wastes material and increases weight	[41], [42], [50]
	Number of chambers	Inner anti-bottoming and multiple bags add margin and redundancy (fail-safe philosophy), but increase weight and complexity	[34], [35], [37], [39], [41], [42]
	Tethers and webbing	Divert inertial loads from the fabric and control the shape, at the cost of added parts	[41], [48], [52]
Fabric	Fabric type	Strength, tear resistance, low permeability, temperature flexibility vs. mass, packability	[41], [42], [45], [47], [48], [49], [52], [53], [54], [55], [56]
	Fabric thickness	Thicker fabric raises the rupture margin but slows deployment, enlarges the package, and increases weight	[39], [46], [57]
Venting	Vent area	Larger orifices dissipate more energy and limit rebound, but risk bottoming-out	[36], [37], [39]
	Vent location	Sets the outflow and whether the ground or payload blocks the vent during the stroke	[41]
	Vent burst pressure	Early opening lowers the peak pressure and resultant acceleration but spends stroke; late opening raises the peak pressure and acceleration	[37], [49]
	Fabric porosity	Distributed gas leakage dissipates energy without vents but reduces pressure retention	[45], [46], [57], [58], [59]
	Vent actuation	Active valves adapt to the actual impact; passive valves need no sensing or power	[32], [33], [34], [40], [41], [42]
Gas	Initial bag pressure	Higher pressure stops the payload in a shorter stroke but raises membrane loads	[37], [39], [49]
	Gas composition	Specific-heat ratio sets compression stiffness; mixtures cut temperature sensitivity	[41], [47], [49], [53], [60]

continued on next page

Group	Variable	Trade-off	Sources
Deployment and packaging			
Inflation	Inflator type	Pyrotechnic vs. stored compressed gas: drives gas output, package size and triggering mechanism	[41], [49], [53], [60], [61]
	Inflator supply pressure	Faster filling vs. higher deployment loads; the flow chokes at sonic conditions	[49], [58]
	Inflator mass flow rate	Shorter injection time vs. higher deployment loads	[46], [57], [58]
	Inflator gas temperature	More pressure per unit gas mass vs. thermal load on the fabric just after inflation	[41], [45], [46], [57]
Housing	Housing size and geometry	Compact housings save mass and volume but pack the bag tighter	[46], [62]
	Housing material	Plastic saves mass over metal; metal bases give a stiff mounting	[38], [46], [53], [62]
Folding	Folding pattern	Governs unfolding kinematics, deployment time and pre-stress in the packed fabric	[45], [55], [57], [58], [62], [63], [64], [65], [66]
	Number of folds	Smaller package vs. longer, less predictable unfolding	RIVEDI LIBRI DI MATEMATICA DELLA CERAGIOLI
Seams	Seam locations	Seams concentrate stress and are the main failure site; patterning minimises their number	[45], [52], [67]

The table conveys the complexity of the design problem at hand, highlighting the large number of design variables. Multi-variable problems typically require an optimisation process where an optimal combination of variable values is achieved through guided iterations based on penalty concepts. This consideration forms the basis for the following chapters, in which the airbag design process is distilled into an ad-hoc workflow tailored for a specific payload application and impact scenario, focusing on a single design phase among the two presented. It is now possible to define the research-development objectives of the present thesis, leading to the proposed solution in Chapter 3 and the specific implementation choices presented in Chapter 4.

2.4. Research-Development Objective

Having presented the context in which the research project is allocated and having gathered the necessary design information, it is possible to address the requirements set by the project commissioners:

- Identify and characterise the mechanical allowables of typical medical payloads based on general packaging requirements for UAV-transported medical goods;
- Develop an informed conceptual design of an airbag system capable of deploying during emergency release conditions to absorb impact energy and preserve payload integrity.

These requirements can be summarised in the following Research-Development Objective and Research Questions:

Research-Development Objective

Develop and validate a design workflow to model and simulate an airbag-based protective system for UAV medical payloads using LS-DYNA, with the primary objective to ensure the structural integrity of pharmaceutical materials and their containment during drone failure and emergency payload release, thereby preventing environmental dispersion and supporting certification requirements.

Research Question(s)

How can a systematic multi-phase simulation workflow be developed to design an airbag-based protective system for medical payloads during UAV emergency-release events?

1. What are the allowable mechanical limits (acceleration, stress, etc.) for medical products during emergency release, and how should these define the design envelope?
2. How can the inflated airbag configuration be optimised to satisfy impact-absorption requirements while balancing competing design requirements?
3. How can the optimised inflated geometry be transformed into a compact and functionally deployable system?

3

Methodology: the Airbag Design Workflow

3.1. Problem statement

To properly outline and, more importantly, test the workflow proposed as a main objective of this thesis, it is necessary to identify the specific context in which this project is embedded, as it originates from prior research work conducted within the specific framework of Turin. The design inputs presented below were gathered from analysing the results presented by the group ([9], [10], [11], [12], and [13]) and categorising them into mission aspects on which to root the proposal of Section 3.2.

3.1.1. Reference mission and design inputs

Route and flight profile

The reference mission is based on a hospital-to-hospital route as initially suggested by Cavallo (2020) [9], connecting two hospitals across the city of Turin as shown in Fig. 3.1, representing a city map with route schematics. In particular, the starting point is the Molinette Hospital in the southern region of the city, and the destination is the San Giovanni Bosco hospital in the northern sector. As they are located far apart, these two hospitals require medical deliveries to traverse a broad part of the city's territory. The hospitals are connected through two possible routes:

- **Route A**: Initially hypothesised by Cavallo and confirmed by Bertelli (2021) [10], it takes advantage of the above-water air corridor over the river Po and its tributary Stura, mostly skirting the city and flying over city ground only in the relatively short segments connecting the hospitals to the rivers. Among the two routes, it is the safest (and, thus, most certifiable according to Cavallo) as it reduces the probability of ground impact in case of drone failure due to the UAV's permanence in the above-water air corridor. This route was selected as viable within the Concept of Operations (ConOps) that informed the preliminary drafting of the SORA. Approximate length: 10 km. Approximate duration time: 15 min, comparable to ambulance-based delivery times.
- **Route B**: A crude alternative to Cavallo and Bertelli's Route A, connecting the two hospitals in a straight line over the city. Angelini (2023) [13] deemed this route infeasible based on their comparative safety analysis of delivery means, primarily due to the lack of proper risk mitigation measures concerning in-flight failures. However, as the approach proposed in this thesis aims at partially mitigating these risks, it was chosen to consider this route (at least for preliminary evaluation).
- **Alternative routes *(omitted from analysis)***: Fiorentino (2021) [12] and Angelini (2024)'s [13] work is rooted in minimum risk-based route selection considering territories that extend beyond city borders, specifically including the entire province[1] in case of Angelini's study. Either using the

[1] In Italy, local government is organised across three administrative levels: regions, provinces, and municipalities. Here, assessing the Province of Turin corresponds to evaluating the territorial sub-region of Piedmont administered by the provincial

RRT* algorithm in Fiorentino's case or the Dijkstra and A* algorithms in Angelini's case, these two studies share a common approach of considering multiple layers (such as population density, shelters, and critical infrastructure CHECK) to calculate the probabilistic minimum-risk paths for different means of transportation (eVTOL, UAV, etc.), without predefined start and end points, thereby determining the optimal drone hub location across the province. valid for FIORENTINO TOO??? The results of these works were omitted from the definition of the reference mission for two main reasons. First, they do not specifically target hospital-to-hospital operations; their scope focuses on optimising general logistics rather than facilitating inter-hospital transport, contrarily to one of the primary objectives of the present thesis. Second, their findings indicate that drone operations, when evaluated against the merit indices adopted in their analyses, are deemed unviable due to safety constraints. In contrast, the objective of this research is precisely to identify a mission scenario that is unfeasible under current conditions and implement risk-mitigation strategies to ultimately enable drone transit.

In either Route A or B, as specified in Bertelli's ConOps, the cruising height is about 75 m, at 17 m/s cruising velocity (approximately 70% of the maximum cruising speed allowed by the UAV platform of 25 m/s, unsustainable for the entire duration of the mission at the time of evaluation), with possible speed variations according to mission urgency if specified in the ConOps.

Accounting for the presence of wind in the operating environment, wind speed in Turin averages about 1.8 m/s (6.5 km/h) [68], with a peak of about 28.6 m/s (103 km/h) in 2013 [69].

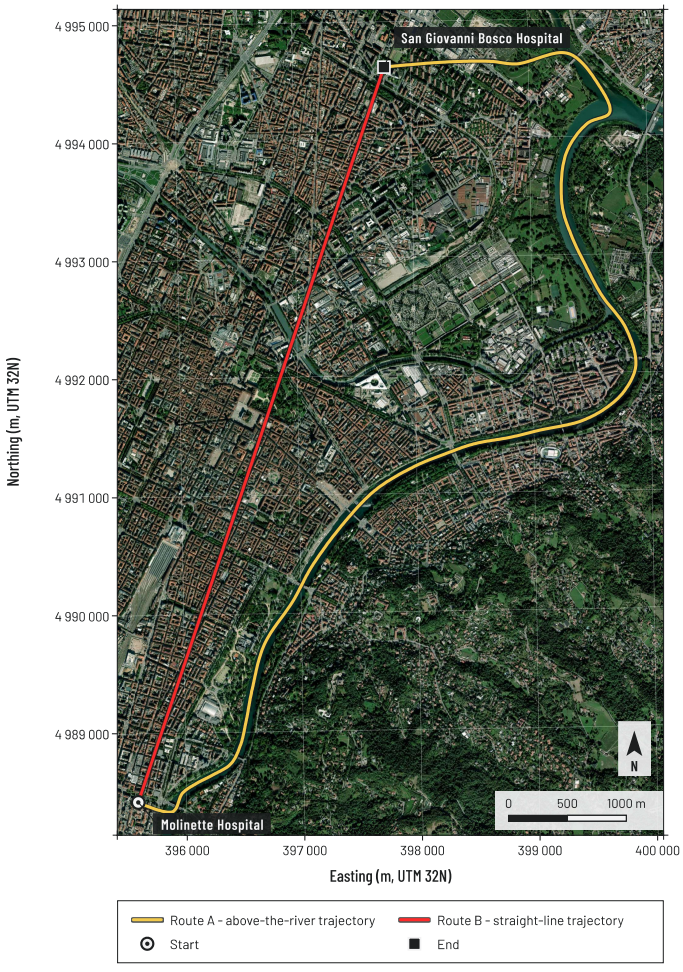


Figure 3.1: Reference mission routes across Turin, from the Molinette Hospital to the San Giovanni Bosco Hospital: Route A (above-the-river trajectory) and Route B (straight-line trajectory).[2]

capital of Turin.

UAV Platform

The UAV platform choice is inherently linked to the reference mission specifications above, meaning that the platform is required to sustain the cruising speed requirements specified in the ConOps. A Maximum Takeoff Mass (MTOM) of 9 kg and a 0.81 m characteristic dimension were also specified in Bertelli's ConOps, leading to the suggestion of a potential drone model suitable for the situation: DJI's *Matrice 300 RTK*. Despite having characteristics suitable for the speed requirements, the drone has a payload limit of 2.7 kg and a wind resistance of 15 m/s in cruise. While this drone was initially selected for the preliminary SORA, its specific parameters are not binding for the present study. In particular, the payload-mass and wind-resistance limitations may be disregarded, given that the ConOps set a much broader requirement for MTOM. In addition to these requirements, Cavallo, in their UAV platform market evaluation, considered several other factors: maximum flight time (usually corresponding to half the autonomy), geometry, glide speed, and frontal area, the latter covering an important role in free-fall drag calculations. By ConOps, the platform is equipped with a parachute, whose contribution is ignored in the calculations regarding the free-fall event.

Failure scenario and impact conditions

The chosen failure scenario implies an in-flight malfunction or fatal failure of the UAV in combination with a failure of the parachute mounted on the drone by ConOps. Excluding the nominal, non-conservative parachute descent of 3 m/s estimated by Bertelli, in this extreme failure scenario the payload is dropped in free air, separated from the drone at an altitude of 75 m above ground. After the failure, a *ballistic* descent mode is chosen for calculations of impact time and point depending on the release position on the map, including the effects of gravity and aerodynamic drag (as opposed to *uncontrolled glide* descent mode, which is applicable to the UAV platform but not to its detached payload), used in Angelini's impact probability study for ground risk evaluation and inspired by PRIMATESTA' CHECK!!!. Wind enters the calculations (as mentioned in the reference route definition) as a horizontal velocity component.

Payload container

The design of the payload container to be carried by the UAV platform in delivery missions was proposed, but never fully prototyped, by Sivieri (2021) [11]. Its geometry is shown in Fig. INSERT FIGURE HERE: The central body is a cylinder capped with two hemispherical domes. In Sivieri's original design, this container had to be constructed via stratifying four layers, from inside out: first layer of 3d-printed PET-G with thickness 2 mm, second layer of manually applied thermal insulation of unknown effective thickness, third layer of 3D-printed PET-G and 2 mm thickness, and a final layer of manually-applied thermo-reflective paint. This sandwich-type structure was designed specifically for evaluating the thermal and vibration loads of a reference mission similar to the one presented earlier, and thus avoiding making considerations on transportability, attachment mechanisms, and other application-specific parameters.

Due to the artisanal nature of this design, linked to the difficulty in modelling specific 3D-printed PETG and the unquantifiable specifications of the insulation material, the payload container was modelled by retaining its shape with a 5 mm thick shell [verify thickness in the FE model] of injection-moulded Acrylonitrile butadiene styrene (ABS), commonly used in packaging. The cavity is filled with polyurethane foam, chosen to simulate a protective wrapping (such as bubble wrap) that medical operators would use as established praxis to mitigate vibrations. The thermal paint is omitted from the payload container modelling process.

Payload

Considering the wide variety of medical goods potentially suitable for UAV-based air transport, it was necessary to identify a payload representative of standard shipments, excluding human organs and specialized equipment (e.g. defibrillators) to focus specifically on pharmaceuticals. Therefore, glass phials of the medicinal product *Akineton Fiale*, containing *biperidene lactate* at a concentration of 5 mg/ml in injectable solution and shown in Fig.3.2[3], were chosen. The product is commercially supplied in boxes of five phials, each individually packaged in a plastic blister tray.

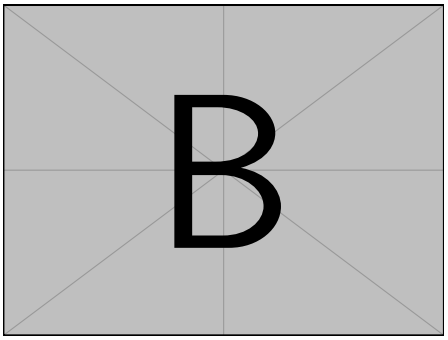
[2]Imagery: Esri, Maxar, Earthstar Geographics, and the GIS User Community. Basemap: Web Mercator; grid: WGS84 / UTM 32N.

[3]Source of photo: `https://www.guttafarm.net/akineton-cena`.

Akineton is a pharmaceutical product commonly used in the treatment of Parkinson's disease and drug-induced extrapyramidal symptoms, and is usually administered by intravenous or intramuscular injection. For the purposes of the present thesis, only the mechanical properties of the glass container are of interest, as the pharmacological action of the medical fluid is not relevant to the type of structural analyses implemented in this study.



(a) Akineton Fiale, 5 mg/ml injectable solution, as supplied in its original packaging.



(b) Single Akineton phial.

Figure 3.2: Akineton Fiale packaging and single phial.

3.1.2. Information gap

The only missing inputs required for a full design workflow evaluation are:

- The derived *impact velocity*, angle, and mass for the selected scenario;
- The quantification of the phials' fragility through the characterisation of their material's mechanical *allowables*;

These failure scenario data must be fed into the design loop(s) proposed below to enable an effective design of the airbag-based impact attenuation system.

3.2. Proposed approach

3.2.1. Outer design loop: formalising the Safety-First principle

In response to the gap identified in Section 2.2, the first cotribution of this thesis is to propose an alternative design approach adopting (and later testing) the following guidelines:

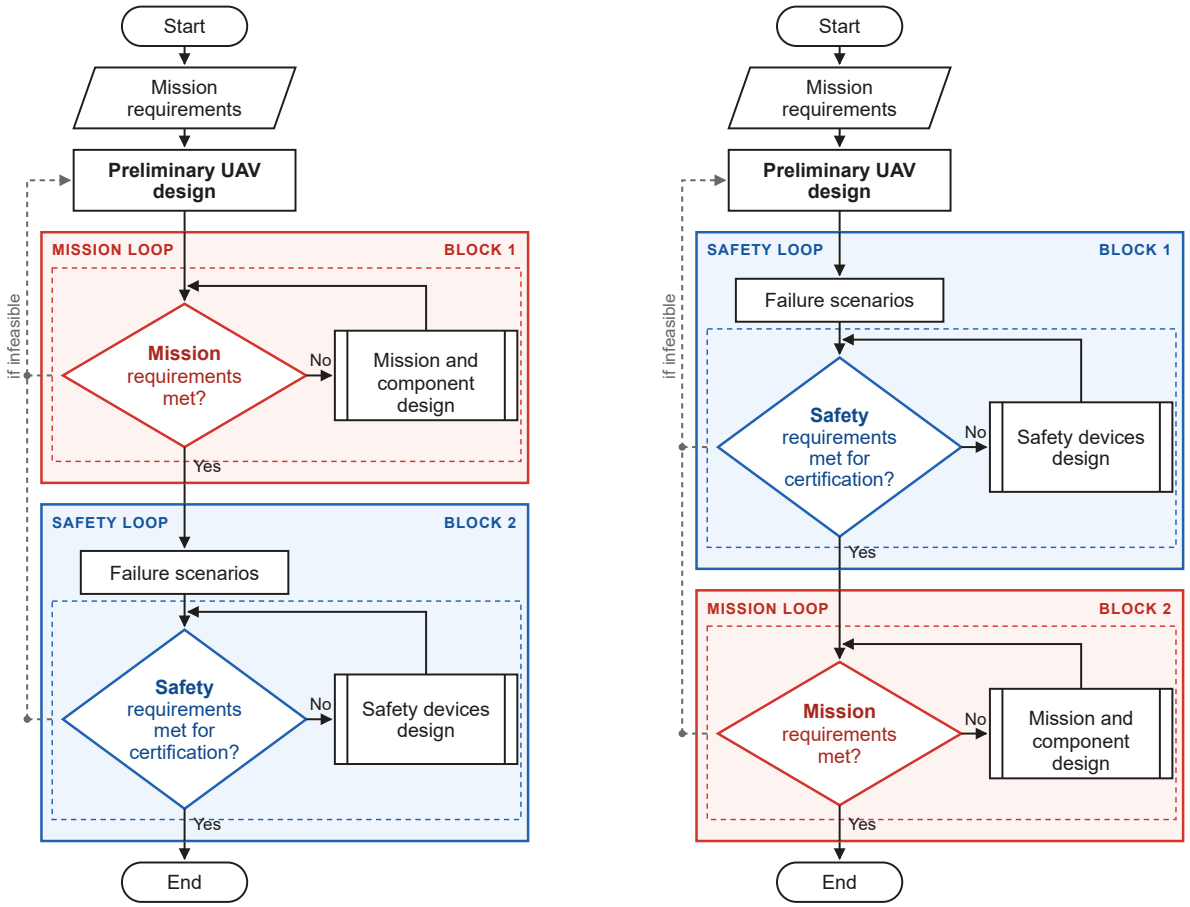
Safety-First Design Principle

1. The airframe and payload must be considered as two distinct entities;
2. In the type of operations considered, saving the payload takes higher priority than saving the airframe (carrying organs, expensive vaccines, emergency medicines, or high-value medical equipment is worth more than the drone itself);
3. Overall, the objectives are to minimise or mitigate:
    - ground impact consequences,

- payload damage (preserving its integrity),
- environmental dispersion;

4. These considerations should be applied at the beginning of the design process, possibly by integrating multiple safety devices (e.g., an airbag and a parachute working in tandem) and adjusting flight-mechanics considerations accordingly.

A direct consequence of applying the Safety-First design principle in UAV-based operations' design is changing the sequence of design events, as exemplified in the simple flowcharts of Fig. 3.3, where Fig. 3.3a and Fig. 3.3b compare scenarios in which the mission loop can be interchanged with the safety loop to achieve design convergence.



(a) *Mission-First* approach

(b) *Safety-First* approach

Figure 3.3: Sequence of design events in the two approaches: *mission-first* and *safety-first* loops.

3.2.2. Inner design loop: a payload-protection subsystem design workflow

This is how this thesis proposes to go through the design process, expanding the "safety loop" highlighted in Section 3.2.1, based on the information distilled in Section 2.3.2. The loop shown in Fig. 3.4 is divided into two main parts. The first part is partially payload-specific and is the deliverable of this master thesis; it extends up to the point where geometry-specific decisions have to be made. The second part leads directly to prototyping, on the basis of more specific design decisions that are tailored to the next step in the outer design loop, the mission loop.

Step 0 - Preliminary mission analysis
This step is about the preliminary analysis of the impact conditions of the payload: it analyzes the conditions in which the pillar is dropped, implements some statistical analysis, and then calculates how the payload arrives on the ground. The useful outputs of this phase are the time of freefall and the impact velocities in the vertical and horizontal directions. mission inputs (t_{fall}, V_x, V_y)

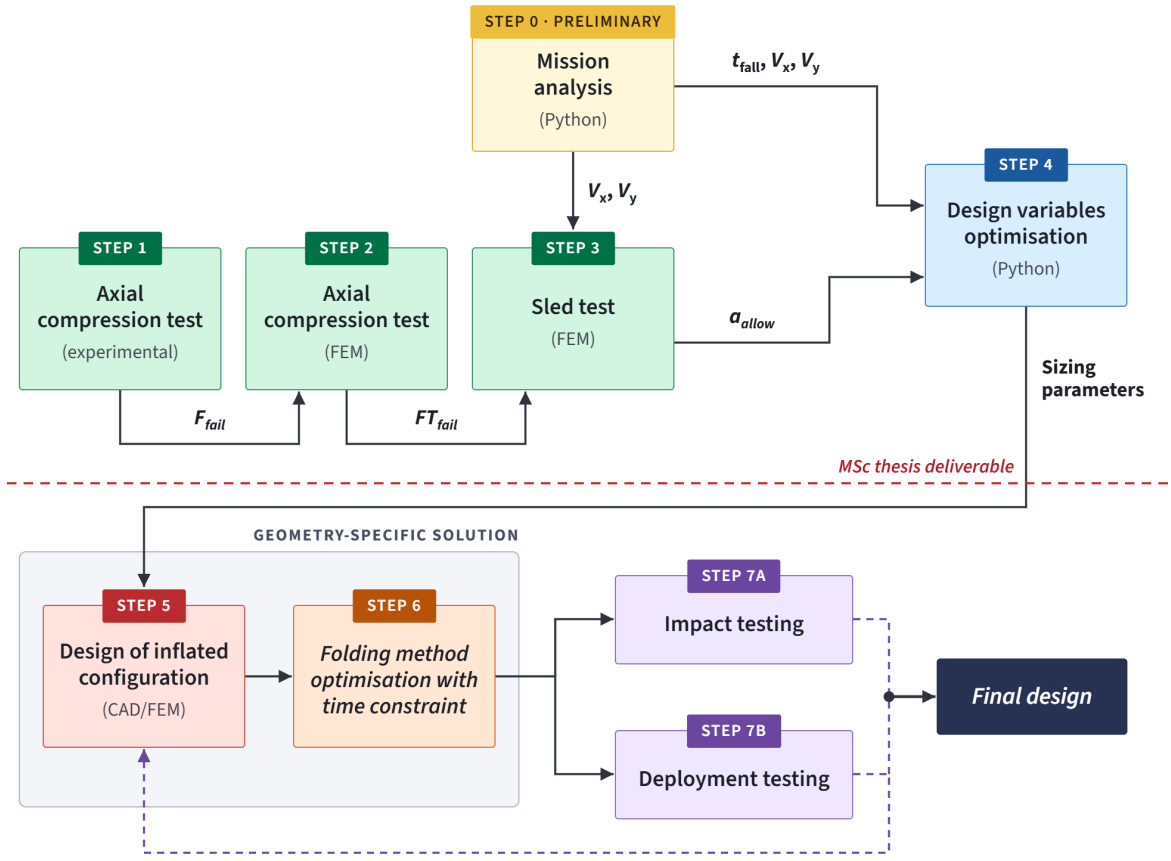


Figure 3.4: Inner design loop, specific for the airbag-based payload-protection subsystem.

Step 1 - Axial compression test (experimental)
experimental characterisation of glass phials breaking point Step 1 is about testing the payload in the lab to obtain information on how it behaves from a breaking perspective. The idea is to conduct a testing campaign to characterize the material the phials are made of, so that this information can be fed to the next step, where this behavior needs to be reproduced at an FEM level. F_{fail}

Step 2 - Axial compression test (FEM)
matching the experimental behaviour at a FEM level to reproduce the mechanical behaviour of the material, limited to the compression behaviour as that is what is going to happen in the event of a fall. To reproduce the behavior observed in the lab, with the objective of creating a material card that can properly characterize the breaking behavior of the phial, the model aims to reproduce the testing conditions as faithfully as possible, in order to produce realistic phials that could then be inserted into a more complete breaking model in the sled test of step 3, FT_{fail}

Step 3 - Sled test
emulation of a sled test inspired by the automotive field, input-output The sled test is about subjecting the payload container, in its design as proposed by Sierra 2021, to realistic impact conditions and scenarios. In particular, the idea is to reproduce an acceleration profile similar to that experienced by the peeled wooden counter in free fall from the conditions described in step 0. The idea is that the

accelerations to which the payload is subjected are strictly linked with the kinetic energy that needs to be dissipated in the impact event. This latter approach is inspired by the automotive industry, which correlates an input acceleration profile with an output acceleration profile. Output: payload allowable acceleration a_{allow}

Step 4 - Design variables optimisation

code, trying to simulate the impact of the actual thing in python to size the airbag This is the fourth and last step of this master's thesis deliverable, and the idea is that it turns all the boundary conditions, such as breaking mode, peak acceleration, and impact velocities, into a viable airbag design that, based on the work presented in the earlier Section 2.1, is optimized following certain criteria like mass, rebound, et cetera, and outputs sizing parameters that can be used to inform the design of a future, geometry-specific design.

Steps 5-7 - Design of inflated configuration, Folding method optimisation with time constraint, Impact and Deployment testing

tailoring the airbag sizing to the payload and UAV geometry matching the gas generator mechanism to the geometry of the drone and payload, and to the impact time and deployment requirements and including folds in the idea, like standard folds, etc. validating the prototype through testing in a feedback loop where virtual and practical behaviours match. Steps 5 and 6 are geometry-specific, meaning that Step 5 is the moment where the fully inflated configuration is designed around a payload, whereas Step 6 is where this full configuration is studied and optimized when it comes to folding patterns, seaming patterns and stowage. The two are inherently interconnected, but Step 6 focuses on the optimization of a different parameter, which is the time of deployment, as explained earlier in the variable definition for airbag design. After a folding mechanism and a full design have been hypothesized, the design is tested through a thorough testing campaign divided into two steps, Steps 7A and 7B, which have to do with impact testing and deployment testing. The solutions hypothesized through FEM and through modeling are put to the test by making a prototype and testing the product in the impact conditions it was initially designed for, in a loop between design and testing until convergence is reached and we have a final design ready to be prepared.

4

Results: through the Workflow

Do not include too many implementation details

4.1. Mission and failure scenario analysis: calculating Vx and Vy

* Mi servono degli input: prendo le informazioni raccolte nella fase preliminare di analisi missione (ref analisi missione)per stabilire probabilità di impatto and what not.

* traiettoria * punto di partenza, punto di arrivo * drone scelto * payload scelto * medicinali scelti per simulazione

* serve energia di impatto?

—> risultati del WP2

4.2. Studying the payload: calculating allowable impact acceleration

4.2.1. Material characterisation

TEST

FEM

MENTION MODEL ITERATIONS TRIAL-AND-ERROR IN THE APPENDIX B!!!! FROM ABAQUS IMPLICIT TO CURRENT DECISIONS. Step by step, as in "version 1, abaqus implicit". Version 2, ls-dyna tetra elastic. Etc. MODAL PERTURBATION

* impact of container in LS-Dyna and what not. * diversi angoli di assetto per vedere il caso peggiore di accelerazioni locali * cosa possono sopportare le fialette e il contenitore???? * tipi di impatto, contenitore tipico che le contiene (menzionare la tesi del tipello) e che per il dimensionamento non viene considerato, approssimato a infinitamente rigido e che trasmette le accelerazioni di impatto così come le riceve —-> airbag deve essere dimensionato così ———> tutte le analisi statistiche di prima, che hanno tenuto in considerazione anche il fatto che prima o poi si sorvolerà la città una volta risolti i cavilli burocratici che impongono una certa mitigazione del rischio —> airbag.

—-> risultati del WP3

—-> CONCLUSIONE: I have what I need to design the damn airbag.

4.2.2. Sled test

4.3. Design variables optimisation: the code!!!

```
init: geometry(h=0), m0 = P0·V0/(R·T0)
loop while t < t_max:
  choose dt from previous load (hysteresis)
  F_n = max((P-Pamb)·A_prev, 0) → a_x, a_y          [explicit: old P, old A]
```

```
u_new = u + a·dt ;  h_new = clamp(h + u_n,new·dt)  [semi-implicit Euler]
A, V = geometry(h_new) ;  A_vent = A_or + φ(S - A)
ρ = m/V → P, T adiabatic from previous state
repeat N sub-steps (dt/N):                          [sub-cycling, not iteration]
    if P ≤ Pamb: stop
    dm = ṁ(P,T,A_vent)·dt/N → m, ρ, P (old T), T
P = max(P, Pamb)
if P > P_burst → BURST  | if h ≥ D0 → BOTTOMED | if y_pen ≤ 0 → REBOUND
advance state
```

Table 4.1: Inputs of the preliminary airbag optimisation: fixed inputs and boundary conditions (left) and optimisable design variables (right).

Fixed inputs and boundary conditions		Optimisable design variables	
Impact scenario		**Continuous (searched by the optimiser)**	
M_p	0.5 kg, payload mass	D_0	0.39 m to 1.5 m, diameter
u_0	17.2 m s^{-1}, vertical impact speed	L_0/D_0	0.5 to 4.0, aspect ratio (cylinder only)
$u_{x,0}$	5.0 m s^{-1}, horizontal impact speed	P_0	P_{amb} to 1.4×10^5 Pa, inflation pressure
$u_{n,0}$	≈ 17.9 m s^{-1} *along* $\mathbf{n}$, *at* $\theta \approx 16.2°$ *(derived)*	φ_{vent}	0 to 0.10, vent open-area fraction
Environment		**Discrete – integer (searched by the optimiser)**	
P_{amb}	101 325 Pa, ambient pressure	n_{ply}	$\{1, \ldots, 4\}$, number of fabric plies
g	9.81 m s^{-2}	**Discrete – categorical (enumerated)**	
T_0	288 K, initial gas temperature	Shape	sphere ($k = 0.5$), cylinder ($k = 1$)
Requirements and safety factors		Fabric	Nylon 6,6, silicone-coated: 600 N cm^{-1}, 180 g m^{-2}
a_{allow}	500 m s^{-2}, allowable payload acceleration		PET, PU-coated: 550 N cm^{-1}, 190 g m^{-2}
SF	1.2, safety factor on a_{allow}		Aramid, coated: 1000 N cm^{-1}, 260 g m^{-2}
a_{lim}	$a_{allow}/SF \approx 417$ m s^{-2} *(derived)*	Gas	N_2: $R = 296.8$ J kg^{-1} K^{-1}, $\gamma = 1.400$
$u_{res,max}$	2.0 m s^{-1}, residual speed cap		He: $R = 2076.9$ J kg^{-1} K^{-1}, $\gamma = 1.667$
η_{seam}	0.70, seam efficiency on fabric strength		Ar: $R = 208.1$ J kg^{-1} K^{-1}, $\gamma = 1.667$
SF_{burst}	1.50, design factor on membrane tension		CO_2: $R = 188.9$ J kg^{-1} K^{-1}, $\gamma = 1.289$
PEN	10^4, penalty weight on constraint violations		Dry air: $R = 287.0$ J kg^{-1} K^{-1}, $\gamma = 1.400$

* codice ispirato da Zhou et al 2019 - spiegazione del modello e dell'aspetto dei vari blocchi * Sapendo che le thresholds delle provette sono tot, allora si può convergere verso un pre-dimensionamento dell'airbag con tot dimensioni o caratteristiche. * oracle code for direzionamento verso un punto di partenza per il design

* I CHOSE THE CRUDE IMPLEMENTATION — THIS IS BECAUSE THE ONLY COUPLING ANALYTICAL MODEL IS MISSING AND I CANNOT RETRIEVE IT. THEREFORE THIS IS MY IMPLEMENTATION XXX Title: "Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers"

Author: Timothy F. Patterson

Report: NATICK/TR-87/042 (DTIC accession ADA199593)

Date: August 1985 (work performed March 1984–May 1985)

4.4. Summary of results

Actual design variables p inflation, diameter, shape, etc

RESULTS: BRING IT BACK TO THE RESEARCH QUESTIONS!!!!!

5

Conclusions

5.1. General summary

Bring back the idea of the safety-first workflow into the broader context of the long-term, drone-network-all-over-Italy vision.

5.2. Recommendations for future work

Feeding these results in the "outer" design loop, updating the hardware and mission design, re-evaluating initial conditions to re-inform the "inner" design loop -> once reached convergence, the need to simulate and test the design implemented on the drone, with informed decisions on inflation mechanism and folding method.

combination of parachute (for, let's say, half the falling height) and then airbag for the second half? the objective is to minimise oveall mass.

we are not done at all. There is the geometry-specific part that is required, including the design recommendations in Section 2.3.2 and going on, PAST the workflow, as mentioned in Section 3.2.2

6

Project Management

Inspired by the Gibb's reflective cycle (?)

6.1. Strategy and planning

planning strategy (especially at the beginning), 5 supervisors to update independently, different means of updates - weekly meetings that became only when i had updates - regular bi-weekly meetings - irregular meetings and informal discussions here and there because of huge workload

6.2. Challenges

6.2.1. challenge 1: aligning interests for thesis scope

- 5 supervisors with 5 different visions of my thesis work. what contributed was a lack of a unified platform to check my progress and congregate materials.

6.2.2. chellenge 2: Computational resources NO MORE

- by contract, politecnico should have given the resources, hpc and licences - unexpected maintenance, with delay - Bianca Ex Machina, access to DelftBlue. Out of contract, otherwise the thesis should have been changed

6.2.3. challenge 3: IT infrastructure bottleneck

- resulting project flow from company pc to personal pc, because no permission to install a VPN and launch from the company's computer.

6.2.4. challenge 4: going against the judgement of supervisors

- scegliere shell invece che solids in case of fev supervisor - scegliere full invece che reduced model in case of tudelft supervisor.

6.3. Lessons learned

I'm really mad

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A

Extended Market Analysis

A.1. Value-chain actors

Table A.1: Actors involved in drone delivery of medical goods in Italy, grouped by value-chain stage. † Based in Piedmont.

Category	Institution	Scope
Drone manufacturers	SkyEU†	Autonomous high-speed eVTOL drones for first aid and critical logistics [27]
	FlyingBasket	Cargo drones; used for blood transport Pozzuoli–Ischia [70]
	H2C	Hydrogen drones and hydrogen supply system for Padova Sandbox [71]
	Dronus	"Radon" drone for medical delivery [72]
	Eurolink Systems	"Beluga" medical delivery drone, with Università Cusano [72]
	IDS Ingegneria Dei Sistemi	Defibrillator-carrying drone (WingBeat project) [73]
Technology partners (role not specified)	Mavtech	INDOOR technical partner; aerial surveillance and operational-support products [18]
	FEV Italia†	Industrial partner of the DIMEAS feasibility studies; numerical impact analysis
	Leonardo	VTOL drone demonstration flights in Lazio, with Telespazio and D-Flight [74]
	Caltec	Technical partner of SIS 118 for the SEUAM defibrillator drone [75]
	AP 74 UAV	Drone company involved in the Piadena pharmacy link [76]
Payload / container makers	ABzero	Smart Capsule: sensorised, UN3373-certified container for blood, organs and drugs [21]
	Carpitech	"Philotea" temperature- and atmosphere-controlled capsule [77]
	Air Abruzzo	"Flying Box" transport capsule [77]
	Telespazio	TPZ Air 100 drug box used in U-ELCOME [78]
Operators	ABzero	Drone operator for San Raffaele trials; handled ENAC authorisation for INDOOR [23], [79]
	Unmanned4you	Operated Aetos blood-transport flights with FlyingBasket [70]
	DL Droni	Piloted the Piadena pharmacy flights [76]
	Telespazio	Runs the U-ELCOME hospital-to-village drug delivery service [16]
	Croce Rossa Italiana	Own drone fleet and certified pilot school; urgent-drug delivery trials [80], [81]
Airspace, infrastructure and energy	ENAV	Air navigation service provider; coordinated Corus-XUAM [82]
	D-Flight	U-space services (ENAV group); Corus-XUAM partner [82]
	DTA	Corus-XUAM partner, Grottaglie airport test site [82]

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Category	Institution	Scope
	ADPM Drones	"Hangar": docking and recharging station for automated missions [82]
	Gruppo SAVE	Airport group leading Padova Sandbox [71]
	Heron Air	Operator of Padova airport, Sandbox partner [71]
	Milani; DBA Group	Build the hydrogen production and refuelling unit for Padova [83]
	Stesi	Supervision system, ground logistics and spare-parts analytics for Sandbox [84]
	Torino Caselle and Cuneo Levaldigi airports†	Proposed hub and support base for hydrogen drone logistics [24]
Research	PIC4SeR, Politecnico di Torino†	Autonomous flight technologies for transporting biological material (INDOOR) [20]
	DIMEAS, Politecnico di Torino (Prof. E. Cestino)†	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost–safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Università di Torino†	INDOOR project partner [18]
	Scuola Superiore Sant'Anna	Parent institution of the ABzero spin-off [23]
	Università Cusano	Co-developer of the "Beluga" drone [72]
	San Raffaele technology centre	Developed the FF2020 drone delivery demonstrations [85]
End users	AOU Città della Salute e della Scienza†	INDOOR test flight between hospital sites; proposed hydrogen logistics plan; pharmacy department partner of DIMEAS feasibility studies [22], [24]
	Regional Transplant Centre Piemonte–Valle d'Aosta†	INDOOR partner for organ and biological transport [18]
	Ospedale San Raffaele	In-campus delivery of drugs and biological samples [86]
	ASST Sette Laghi	Feasibility study for inter-hospital drug transport [87]
	Piadena community pharmacies	Pharmacy-to-pharmacy drone link [76]
	ASL 1 Abruzzo	Hospital-only drugs delivered to remote villages [16]
	ASL Napoli 2	Blood transport to Ischia [70]
	Ospedale Miulli	Demonstration flight of life-saving drugs [88]
	ASL Lecce	Planned Gallipoli–Casarano hospital link [89]
	Ospedale di Patti	Planned link to the Aeolian Islands [89]
	Ospedale Pediatrico Bambino Gesù	Partner in VTOL demonstration flights in Lazio [74]
	SIS 118	Emergency service promoting the SEUAM defibrillator drone [75]
Promoters and funders	Fondazione DOT†	Promoter of the INDOOR project [18]
	Centro Nazionale Trapianti	Funds the INDOOR research grant at PIC4SeR [19]
	City of Turin†	INDOOR partner; funded the LastMed call [18], [25]
	Regione Veneto	Signatory of the Padova Sandbox agreement [71]
	Regione Lazio	Partner in VTOL demonstration flights [74]
	Regione Campania	Institutional support for the Aetos trial [90]
	EU research programmes	H2020 Flying Forward 2020; U-ELCOME, co-funded by Horizon [79], [91]
	Abruzzo municipalities and Parco Sirente Velino	Host communities of the U-ELCOME service [16], [78]
Regulators and authorities	ENAC	National civil aviation authority; flight authorisations; protocol with the Red Cross [92]
	EASA	EU rules for drone operations, Regulation (EU) 2019/947 [93]
	Regione Piemonte†	Regional health authority; drone trials subject to benefit–risk analysis [28]
Alternatives	Tecnocad / Tecnomill / Bylogix†	LastMed autonomous ground rover for last-mile drug delivery [26]

A.2. Trials and planned initiatives

Table A.2: Drone delivery of medical goods in Italy: trials and planned initiatives (status as reported by the cited sources; checked 5 October 2026).

Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
Piedmont	Turin, Molinette ↔ San Giovanni Bosco feasibility studies	2020–2024	DIMEAS (Politecnico di Torino); AOU Città della Salute e della Scienza (Pharmacy); FEV Italia	Life-saving drugs, blood-derived medication	Approx. 10 km corridor along the Po, approx. 15 min at 17 m/s	Desk studies (5 MSc theses, 1 conference paper); no flights performed	[9], [10], [11], [12], [13], [14]
	Turin, INDOOR	Oct 2024	Fondazione DOT; PIC4SeR (Politecnico di Torino); ABzero; AOU Città della Salute e della Scienza	Biological samples (blood) in sensorised capsule	CTO → Molinette, 500 m, automatic navigation over the city and the Po	Single experimental flight	[18], [22]
	Turin, hydrogen drone logistics plan	Apr 2025	AOU Città della Salute e della Scienza (Commissioner T. Schael)	Drugs, small medical loads	Torino Caselle hub, Cuneo Levaldigi support, future Parco della Salute	Proposal; no implementation found	[24], [94]
Abruzzo	Avezzano → Collelongo, U-ELCOME	Jul 2025	Telespazio (Leonardo/Thales); ASL 1 Abruzzo; municipalities of Collelongo, Montereale, Rocca di Mezzo	Hospital-only drugs for frail patients	Hospital to village pharmacy, approx. 20 min	Start of operational phase; further routes announced	[16], [17]
Apulia	Grottaglie → Manduria, Corus-XUAM	by 2022	ENAV; D-Flight; DTA	Medical material	Airport to hospital	Test (reported in 2022)	[82]
	Taranto, SEUAM	Feb 2024	SIS 118; Caltec (Campanian aerospace consortium)	Automated external defibrillator	Urban, simulated cardiac arrest	First test of experimental project	[75]
	Lecce, ASL Lecce	2025	ASL Lecce	Not specified in source	Gallipoli–Casarano hospitals	Intended	[89]
	Acquaviva delle Fonti, Ospedale Miulli	Nov 2025	Ospedale Miulli	Life-saving drugs, biomedical material	25 kg hexacopter; 25 km in 25 min	Operational demonstration	[88]
Campania	Pozzuoli → Ischia, Aetos	Feb 2025	FlyingBasket; Unmanned4you; ASL Napoli 2	Blood, biological samples	17 km in 15 min; 4 flights; 5 kg carried	Second trial phase	[70], [95]
Lazio	Rome	2022	Leonardo; Telespazio; D-Flight; Ospedale Pediatrico Bambino Gesù; Regione Lazio; Aeroporti di Roma	Not specified in source	Fixed-wing VTOL drones	Two demonstration flights	[74]
Lombardy	Piadena (Cremona)	2021	Two community pharmacies; DL Droni; AP 74 UAV	Drugs and devices between pharmacies	Approx. 6 kg electric drone, approx. 1 kg payload	Announced as continuous service; current status not verified	[76]

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Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
	Milan, Ospedale San Raffaele, H2020 Flying Forward 2020	Mar 2023	Ospedale San Raffaele; EuroUSC Italia (regulatory consultant); ABzero (operator)	Drugs, biological samples	Within the hospital campus	Demonstration campaign, each flight ENAC-authorised	[79], [86]
	Varese, ASST Sette Laghi	2025	ASST Sette Laghi	Drugs, urgent materials	Between two hospitals a few hundred metres apart	Preliminary feasibility study	[87]
Sicily	Trapani, Croce Rossa Italiana	2021	Croce Rossa Italiana	Urgent drugs	Port quay → vessel; 5 min 30 s vs 35 min by sea	Simulation exercise	[81]
	Patti → Aeolian Islands	2025	Ospedale di Patti	Not specified in source	Patti–Lipari	Under development	[89]
Veneto	Padova, Padova Sandbox	from Mar 2025	Gruppo SAVE; ENAC; Regione Veneto; H2C/H2G; Heron Air; Milani; DBA Group; Stesi	Medical goods, biological samples (hydrogen drones)	Up to 100 km, 4–5 kg payload (stated)	Ground tests of hydrogen system; first flight expected autumn 2026	[71], [96]

B

FEM Iterations for glass phial model

Adding source code to your report/thesis is supported with the package `listings`*. An example can be found below. Files can be added using* `\lstinputlisting[language=<language>]{<filename>}`*.*

```
"""
ISA Calculator: import the function, specify the height and it will return a
list in the following format: [Temperature, Density, Pressure, Speed of Sound].
Note that there is no check to see if the maximum altitude is reached.
"""

import math
g0 = 9.80665
R = 287.0
layer1 = [0, 288.15, 101325.0]
alt = [0,11000,20000,32000,47000,51000,71000,86000]
a = [-.0065,0,.0010,.0028,0,-.0028,-.0020]

def atmosphere(h):
    for i in range(0,len(alt)-1):
        if h >= alt[i]:
            layer0 = layer1[:]
            layer1[0] = min(h,alt[i+1])
            if a[i] != 0:
                layer1[1] = layer0[1] + a[i]*(layer1[0]-layer0[0])
                layer1[2] = layer0[2] * (layer1[1]/layer0[1])**(-g0/(a[i]*R))
            else:
                layer1[2] = layer0[2]*math.exp((-g0/(R*layer1[1]))*(layer1[0]-layer0[0]))
    return [layer1[1],layer1[2]/(R*layer1[1]),layer1[2],math.sqrt(1.4*R*layer1[1])]
```

C

Task Division Example

If a task division is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Table C.1: Distribution of the workload

	Task	Student Name(s)
	Summary	
Chapter 1	Introduction	
Chapter 2		
Chapter 3		
Chapter *		
Chapter *	Conclusion	
	Editors	
	CAD and Figures	
	Document Design and Layout	

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